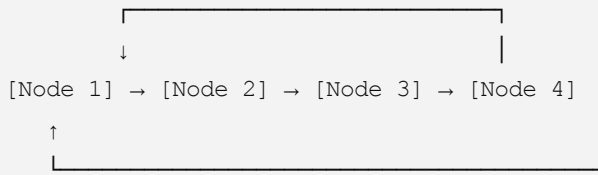


Usage of Circular Linked List

A **Circular Linked List (CLL)** is a linked list in which the **last node points back to the first node**, forming a circle.



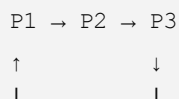
Because there is no **NULL** at the end, we can continuously traverse the list from one node to the next.

Common Applications

Application	How Circular Linked List is Used
1. Round-Robin CPU Scheduling	Processes are arranged in a circle. Each process gets a fixed time slice, then control moves to the next process.
2. Circular Queues	After reaching the last position, the queue can continue from the first position. This makes efficient use of available space.
3. Multiplayer Turn Management	Players are stored in a circular list. After one player's turn, control automatically moves to the next player.
4. Repeated Playlist Systems	Songs can be arranged in a circle so that after the last song, playback automatically returns to the first song.
5. Token-Passing Systems	A token is passed from one node/device to the next, eventually returning to the first node.
6. Josephus Problem	People are arranged in a circle, and every kth person is removed repeatedly until a single person remains.

1. Round-Robin CPU Scheduling

Suppose there are three processes:



With a time quantum of 2 ms:

P1 → P2 → P3 → P1 → P2 → P3 → ...

The circular structure naturally represents the repeated scheduling cycle.

2. Multiplayer Turn Management

For four players:

Player 1 → Player 2 → Player 3 → Player 4



After Player 4's turn, the next turn automatically goes back to Player 1.

3. Repeated Playlist

Song A → Song B → Song C → Song D

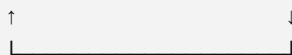


After **Song D**, playback returns to **Song A**.

4. Josephus Problem

If people are arranged as:

1 → 2 → 3 → 4 → 5 → 6



Starting from a particular person, every **k-th person is eliminated**. The circular linked list makes it easy to continue counting after reaching the last person.

Key Advantage

The main advantage of a circular linked list is:

It is suitable for applications where data needs to be processed repeatedly in a cyclic order.

Typical DSA keywords:

Round Robin → CPU scheduling

Players → Turn management

Songs → Playlist

Devices → Token passing

People → Josephus problem

